ECDLP across a large prime conductor gap on a deep 2-isogeny volcano

Study of 2026-10-22, on public synthetic prime-field curves. Classification: theorem note plus a preregistered, controlled measurement (rho against rho). No security statement about any deployed curve.

Question. Is ECDLP hardness the same on both sides of a large prime conductor gap, on a deep isogeny volcano?

Answer

Proved (THEOREMS.md, with Lean for the parts Lean can state):

- **Pairwise, for every algorithm.** For two given curves across the gap, ECDLP on one reduces to ECDLP on the other at the cost of the isogeny chain between them (Theorem 2). In the class measured here every chain is explicit and costs $O(N + d)$ field operations per instance (Theorem 3). In general classes, Galbraith (2024) puts it at $\tilde{O}(q^{1/4})$ (as reported). Either way the best attack cost T^* agrees across the gap up to an additive term below $\sqrt{q}$.
- **Everything that depends on the group alone agrees.**
  - group order and embedding degree;
  - the generic lower bound (Theorem 1).
- **The one exception is a constant, and it travels.** A $j = 0$ or $j = 1728$ crater has extra automorphisms, so rho run *on that curve* is $\sqrt{3}$ or $\sqrt{2}$ cheaper. A cheap ascent carries any floor instance there, so T^* does not split (Theorem 4). This study's class ($D_0 = -11$) has $Aut = \pm 1$ everywhere.

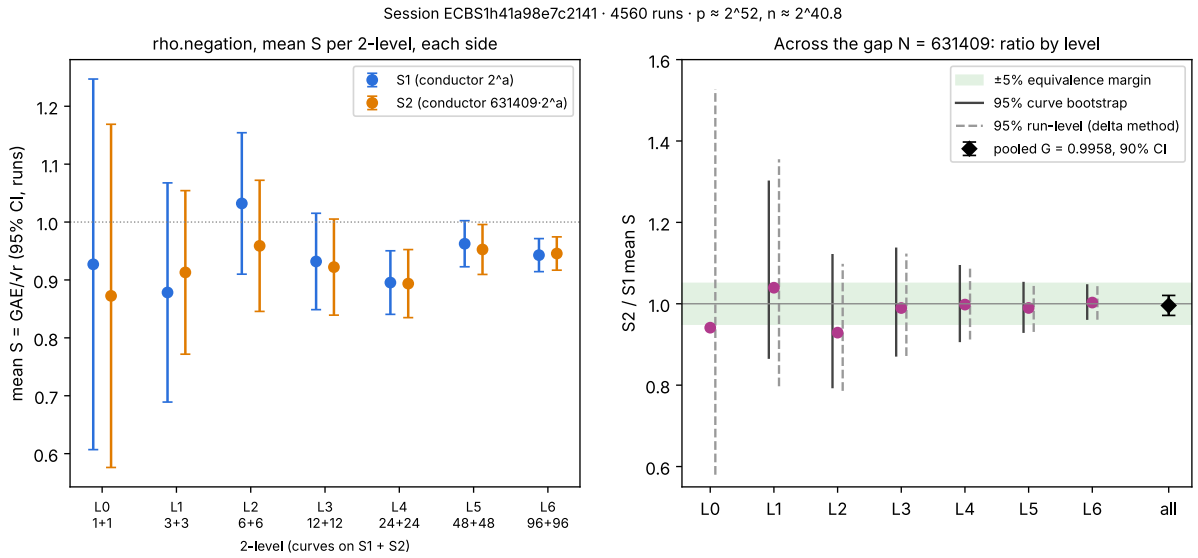
Open. Whether a weakness on a fraction of floor curves can be kept from a given crater curve (KKM 2011, Galbraith's Problem B) is still open in general. In this class it cannot: descending is $O(N)$ too (Theorem 3(g)).

Measured (protocol frozen before data, PROTOCOL.md):

- the class: $p \approx 2^{52}$, gap prime $N = 631409 \approx 77 \cdot p^{1/4}$, a depth-6 2-volcano on each side;
- the method: `rho.negation`, the strongest generic method for $Aut = \pm 1$;
- the coverage: all 380 curves, one ECDLP instance carried across the gap by a verified degree-N Vélu isogeny.

The counted cost is **equivalent within ±5%** across the gap:

	value
G = mean $S(S2)$ / mean $S(S1)$	**0.9958**
90% CI (curve bootstrap, 4,000)	[0.9714, 1.0205]
95% CI	[0.9663, 1.0257]
permutation p (10,000)	0.78
verdict (frozen rule)	**equivalent within ±5%**: the 90% CI is inside [0.95, 1.05] and both planted controls are detected
runs	4,560 (2,280 per side), all verified

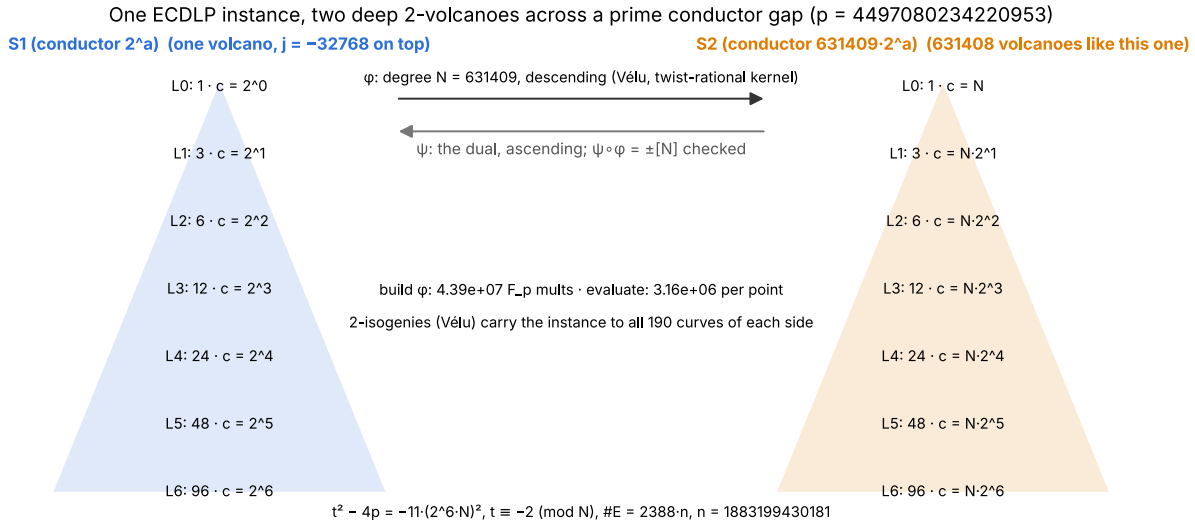


The class and the instance

`volcano-gap construct --bits 52 --depth 6 --d0 11 --max-cofactor 16384 --seed 20261022` (`run/pair.json`):

quantity	value
`p`	4497080234220953
`t`	−5051274, and `t ≡ −2 (mod N)`
`t² − 4p`	`−11 · (2⁶ · 631409)²`
sides	S1: conductor `2^a`, one volcano, crater `j(−11) = −32768`
	S2: conductor `631409·2^a`, `N − 1` volcanoes of the same shape
curves per level (each side)	1, 3, 6, 12, 24, 48, 96 (190); predicted by the class-number formula, walked by `volcano-rho`, and matched by the independent `ca volcano` walker
`#E`	4497080239272228 = 2388 · `n`
`n`	1883199430181 (prime, ≈ 2^40.78)
order proof	`n > 4√p`, so the Hasse interval holds one multiple of `n`
`N`-isogeny φ (S1 crater ⟶ S2 crater)	kernel = a point of order `N` on the quadratic twist; Vélu over `F_p` from 315,704 abscissae
φ checks	homomorphism on 8 random pairs, `φ(P)` of order `n`, codomain order
dual ψ	built from the S2 root's one twist-rational line; reaches `j(−11)`, `ψ∘φ = ±[N]` on 5 points
cost of φ	build: 0.15 s (4.4·10⁷ `F_p` multiplications); evaluate: 3.2·10⁶ per point (0.019 s)
side certificates	all 760 curve models: S1 twists have `(Z/N)²`, S2 twists a cyclic `N`-part, proved by a witness (`run/certificates.json`); no `j` shared

The generator `P` sits on the S1 root and `φ(P)` on the S2 root. `volcano-rho --generator` carries each down its volcano by verified Vélu 2-isogenies (189 edges per side). `planted_shared` targets give the same `k` on all 380 curves, so on model 0 the S2 instance *is* the φ-image of the S1 instance.



Results

Counted unit: `S = GAE/√r`, `r = n` on every curve. Measured unless marked derived.

Per side (`rho-neg`, 2 models × 6 targets per curve):

side	runs	mean S	95% CI (runs)
S1	2280	0.94288	± 0.0203
S2	2280	0.93889	± 0.0208

For reference, the ideal negation-map rho is `√(π/4) = 0.886`. Both sides read about 6% above it, the walk's own overhead.

Per level (S2/S1; level 0 has one curve per side):

level	curves per side	S1 mean S	S2 mean S	S2/S1	95% CI (curve bootstrap)	95% CI (run-level, post hoc)
0	1	0.9270	0.8725	0.941	(one curve: no spread)	[0.580, 1.528]
1	3	0.8784	0.9130	1.039	[0.868, 1.300]	[0.797, 1.355]
2	6	1.0321	0.9589	0.929	[0.795, 1.119]	[0.786, 1.098]
3	12	0.9319	0.9222	0.990	[0.873, 1.135]	[0.872, 1.123]
4	24	0.8955	0.8937	0.998	[0.909, 1.092]	[0.912, 1.092]
5	48	0.9625	0.9526	0.990	[0.931, 1.051]	[0.931, 1.052]
6	96	0.9429	0.9456	1.003	[0.963, 1.045]	[0.961, 1.047]

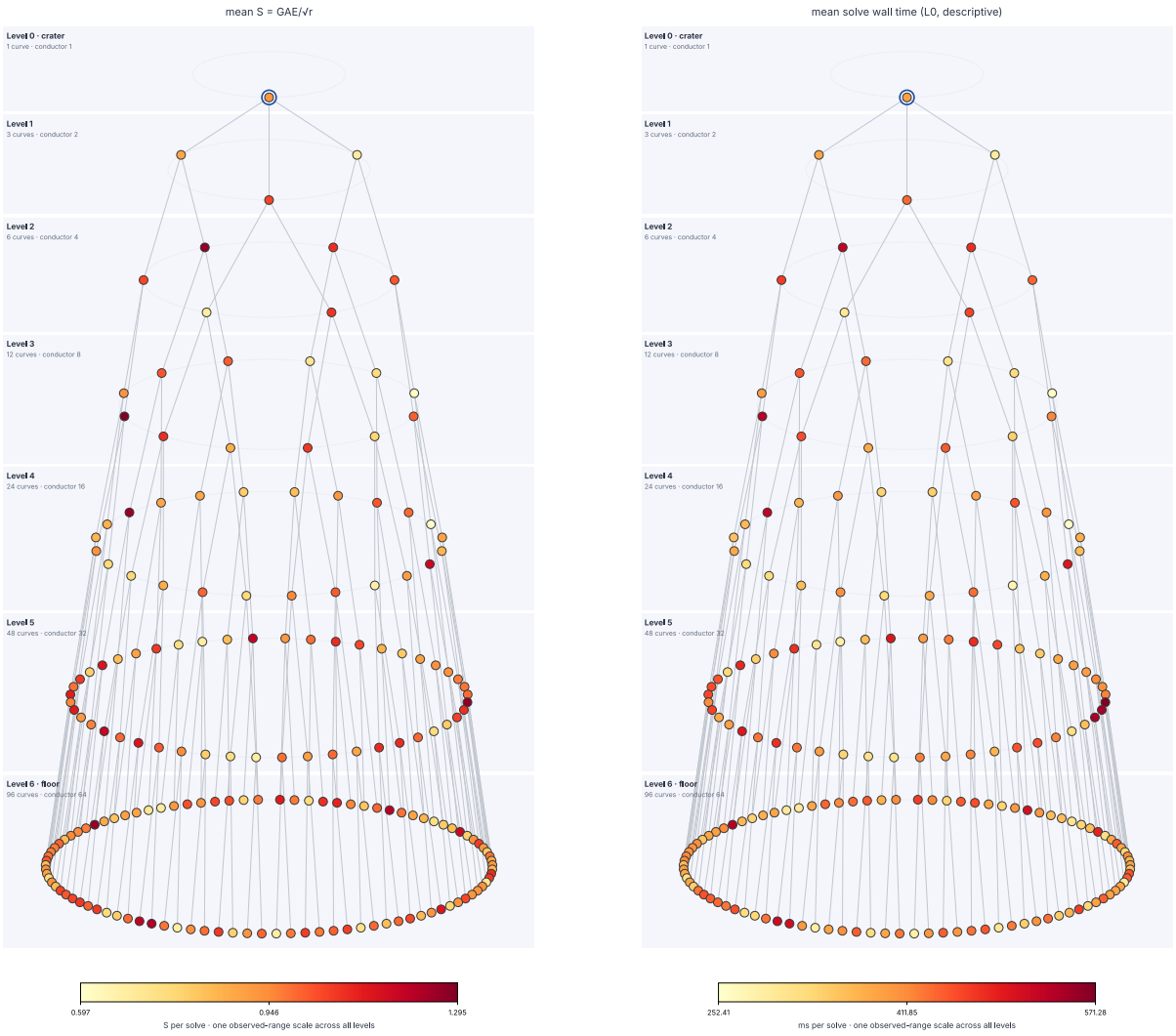
- Every level's interval contains 1.

- The run-level interval (delta method on the two means) was added after the data, because the curve bootstrap has no spread in a one-curve stratum. It is descriptive and does not enter the decision.
- From level 3 down, the two intervals agree. So there is no between-curve variance beyond run noise.

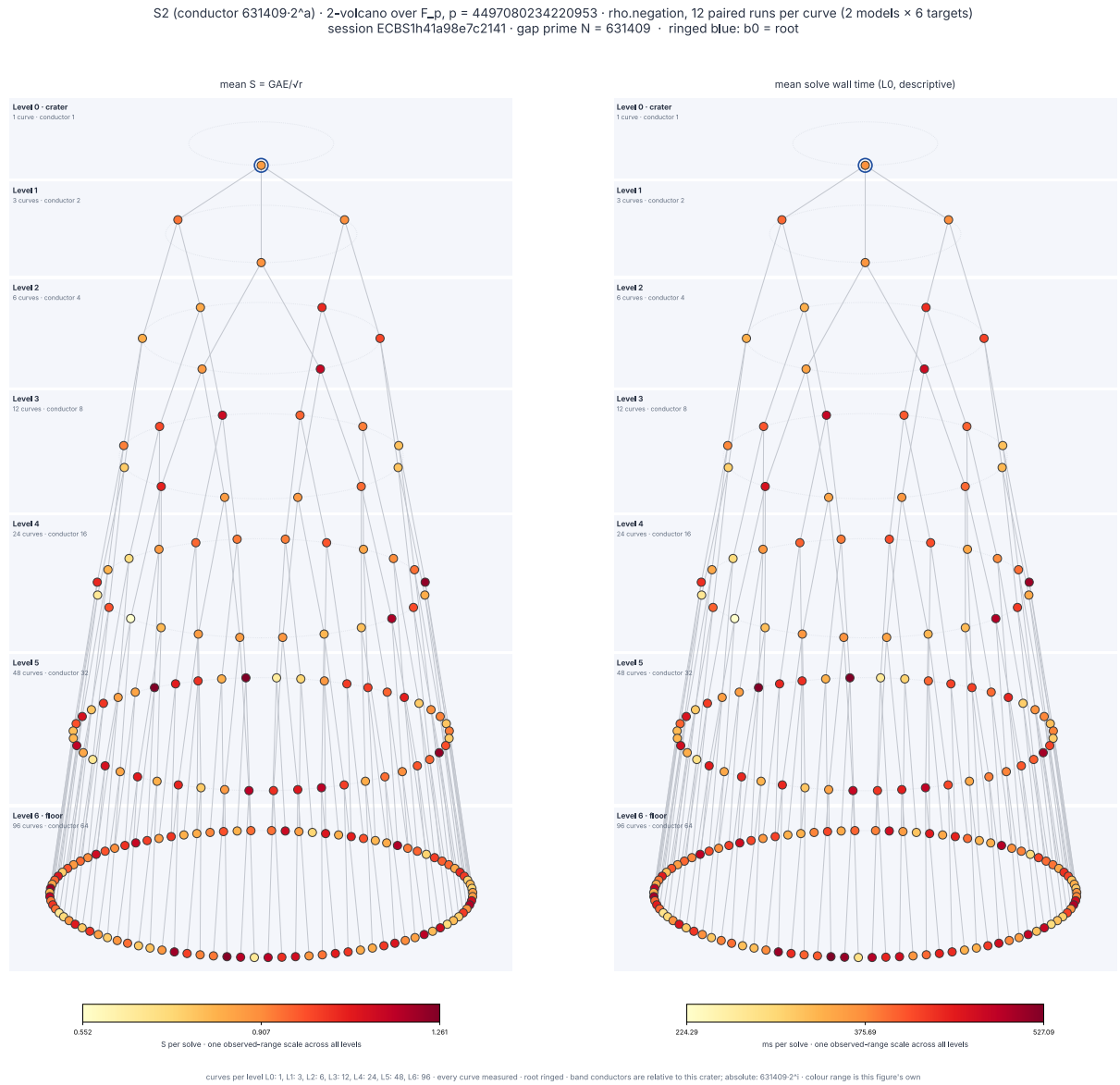
Secondary statistics (all reported in `analysis.json`):

statistic	value	null / reference
crossed bootstrap (shared target draw, conservative)	90% [0.9581, 1.0337], 95% [0.9496, 1.0419]	inside ±5% too
floor/crater within each side (v1.1 stratified)	1.050, 90% [0.889, 1.265], p = 0.65	one crater curve per side: wide
horizontal τ (curves within a level)	6.2%	null p95 9.1%, p = 0.64
model τ (isomorphic models of a curve)	6.2%	null p95 7.8%, p = 0.38
§5a band, S2/S1 at w = 0, 0.1, 1 GAE per unpriced op	0.99576, 0.99576, 0.99576	same method on both sides: reported, not ranked
S at w = 0 / 0.1 / 1	S1: 0.943 / 1.037 / 1.885; S2: 0.939 / 1.033 / 1.877	

S1 (conductor 2^a) · 2-volcano over F_p, p = 4497080234220953 · rho.negation, 12 paired runs per curve (2 models × 6 targets)
session ECBS1h41a98e7c2141 · gap prime N = 631409 · ringed blue: a0 = root



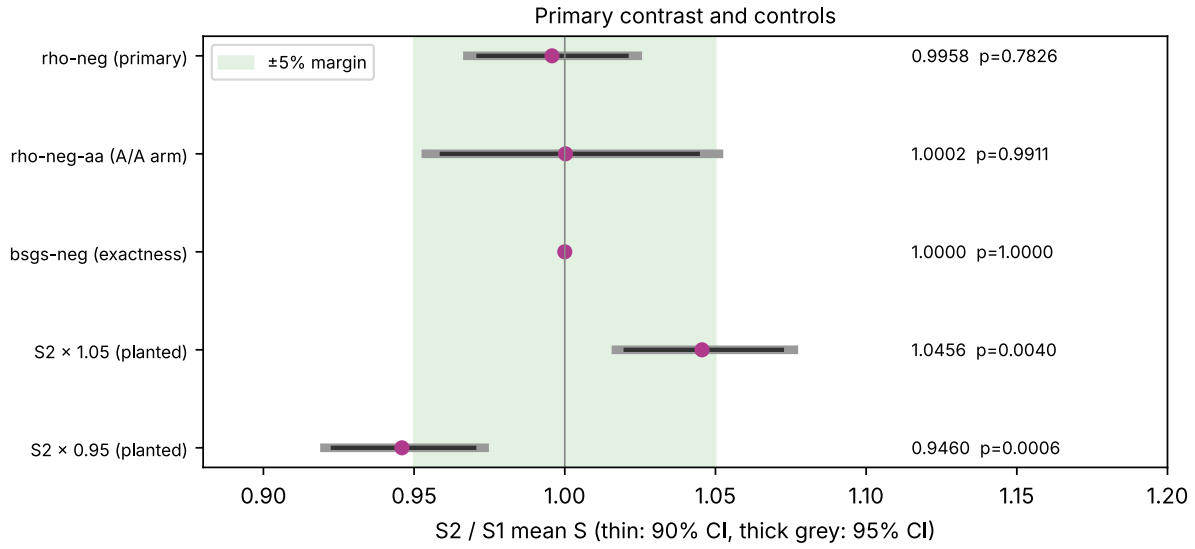
curves per level L0: 1, L1: 3, L2: 6, L3: 12, L4: 24, L5: 48, L6: 96 · every curve measured · root ringed · band conductors are relative to this crater; absolute: 2^i · colour range is this figure's own



Controls

control	prediction	measured	verdict
A/A arm (`rho-neg-aa`, other measurement seed), its own `G`	95% CI contains 1	1.0002, 95% [0.9525, 1.0526], p = 0.99	passes
`rho-neg` against `rho-neg-aa`, matched workloads	1	0.980, 95% [0.944, 1.016]	passes
BSGS exactness (`bsgs-neg`, 2 targets)	one cost per target on all 380 curves, `G = 1`	max distinct cost per target 1; `G` = 1.0000, CI [1.0000, 1.0000]	exact
planted S2 × 1.05	detected	1.0456, 95% [1.0156, 1.0774], p = 0.0040	detected
planted S2 × 0.95	detected	0.9460, 95% [0.9188, 0.9748], p = 0.0006	detected

walk counts	[1, 3, 6, 12, 24, 48, 96] each side	the same from `volcano-rho` and from `ca volcano`	passes
side certificates	all ok, no shared `j`	760/760, 0 shared (re-run by the validator with another seed)	passes
audit	`ecbench verify --replay` 24: exact	24/24 on each of the three sessions, 0 problems	passes



What the measurement says, and what it does not

- **Says (measured).** On this class, `rho.negation`'s mean counted cost is the same on both sides of a prime conductor gap of 631409, to within ±2.9% at 90%. That covers every level of two depth-6 2-volcanoes and one instance carried across by an explicit degree-631409 isogeny.
  - The design can see a 5% shift (both planted controls).
  - BSGS confirms that the instance is the same everywhere.
- **Does not say.**
  - That ECDLP is equally hard for every algorithm. That is Theorem 2's job, with its additive transfer cost.
  - Anything beyond `p ≈ 2^52`, $D_0 = -11$ and this method.
  - The class is special (`t ≡ −2 mod N`), chosen so that the cross-gap isogeny is cheap and explicit. Rho never uses an isogeny, so the special choice does not enter the rho measurement. It does make the KKM isolation question trivial here (Theorem 3(g)).
- **Transfer cost against rho (derived, same host, descriptive).**
  - Carrying one instance across the gap is two evaluations of φ: $6.3 \cdot 10^6$ `F_p` multiplications, about 0.04 s.
  - One `rho.negation` solve averages about $1.29 \cdot 10^6$ group operations ($S \cdot \sqrt{n}$), and 0.39 s of wall time on this host.
  - So the transfer is about a tenth of one rho solve in wall time here. Asymptotically it is `O(N) = O(p^{0.37})` in this family, against $\sqrt{n}$.

Provenance

session	spec	arm	runs	audit

ECBS1h41a98e7c2141	ECS1h7d9934575ec8 (run/specs/spec.json)	rho-neg	4560	audit: 4560 verified, 24 replays exact
ECBS1h4034bfcbccc4	ECS1h0d34427ad5a6 (spec-aa.json)	rho-neg-aa	1520	audit: 1520 verified, 24 exact
ECBS1h658495a369ec	ECS1ha55be72ab53f (spec-bsgs.json)	bsgs-neg	760	audit: 760 verified, 24 exact

- **Host.** Intel Xeon @ 2.10 GHz, 4 vCPU shared VM, isolation L0. The three sessions ran concurrently, one per core. Wall times are descriptive only.
- **Binaries and commit.** ecbench binary sha256 `a59c4014…`. Sessions recorded `git_commit f7b4d94` (the protocol freeze) with `git_dirty: true`; the untracked run directory was in the tree. The ecbench source is unchanged on this branch.
- **Storage.** Sessions are committed with `records.jsonl` and `plan.json` gzipped (`gzip -n -9`). `run/sessions/MANIFEST.sha256` pins the uncompressed bytes. To re-audit, run `gunzip -k run/sessions/*/*.gz`, then `sha256sum -c run/sessions/MANIFEST.sha256` and `ecbench verify`.
- **Lost file.** The `volcano-rho` exports first wrote their `instances.jsonl` to the binary's default `./volcano-rho-out`, where S2 overwrote S1. That file was removed. `run_pipeline.sh` now passes `--out`. The files the study uses (`curves.jsonl`, the specs) are unaffected.
- **Independent checks.**
  - The validator re-ran the replays and recomputed the headline numbers from raw records (`validation/`).
  - `ca volcano` (aburan28/cryptanalysis) re-walked both volcanoes (`run/crosscheck/`): same counts and 189 verified edges per side. It prints no `j`, so curve identities were not compared.
- **Run report.** reports/run-20261022T0844Z/.

Requirement status

requirement	status	evidence
A deep volcano	done	depth-6 2-volcanoes, 190 curves each, every curve measured
A large prime conductor gap	done	`N = 631409 ≈ 77·p^{1/4}`, `log_p N = 0.371` (KKM's regime `p^{1/4} < N < p^{1/2}`)
"Hardness is the same": proof	proved where it can be	Theorems 1–4; Lean: chain transfer and the twist-order arithmetic. The curve-level facts (Vélu, `End`) are checked numerically, not in Lean. The general distributional question (Problem B) is open.
"Hardness is the same": measurement	measured	equivalent within ±5% for `rho.negation`; planted controls detected
Isogeny transport across the gap	done	an explicit degree-`N` map and its dual, verified; cost measured

Validation gate (replay, validator, rho reference)	done	`benchmarks/20261022-deep-volcano-prime-gap/VALIDATION.json`
Index calculus	not run	out of reach at `n ≈ 2^41` on this host; not an IC result, no leaderboard change
Isolated (L2) timing	not run	shared VM; counted units only

Reproduce

See PROTOCOL.md §9. Figures: `python plot.py .` (a scratch venv with matplotlib). It reads `analysis*.json`, `run/pair.json` and the sidecars, and computes nothing. Lean: `cd lean && lake build && lake env lean AxiomAudit.lean`. Tests: `cargo test --release --bin volcano-gap` (one end-to-end test on a 30-bit class).

What can be proved about ECDLP across a large prime conductor gap

Companion to `README.md`. Each statement is labelled **proved here**, **proved in Lean**, **literature (as reported)** or **open**. The Lean file is `lean/DeepGap.lean` it builds on `TransferBound` from `../conductor_gap_cm_pairs_20261010/lean/`.

Setting

- `E/F_p` is ordinary with Frobenius π and trace `t`, and `t² − 4p = D₀·f²` with `D₀` fundamental.
- The `F_p`-isogeny class is `𝓕_t`. Every `E ∈ 𝓕_t` has `End(E) ≅ O_c`, the order of conductor `c = c(E)` in `K = Q(√D₀)`, with `c | f`.
- `N` is a prime with `N ∥ f` (the gap prime). The two sides are `S1 = {E : N ∤ c(E)}` and `S2 = {E : N | c(E)}`. Every isogeny between `S1` and `S2` has degree divisible by `N` (Kohel 1996).
- `n` is a prime dividing `#E(F_p)` with `n > 4√p`, `P` has order `n`, and the instance is `(P, Q = [k]P)`.
- `T_A(E)` is the cost of algorithm `A` on `E`, and `T*(E)` the least cost over all always-correct algorithms.

Theorem 1 — what every curve of the class shares

Status: proved here.

For all `E, E' ∈ 𝓕_t`:

- `#E(F_p) = #E'(F_p) = p + 1 − t`;
- `E(F_p)[n] ≅ E'(F_p)[n] ≅ Z/n`;
- the embedding degree `ord_n(p)` is the same, and so is anomalousness (`#E = p`).

So every attack whose applicability and cost are functions of `(p, #E, n, ord_n p)` applies identically on both sides of any conductor gap. These are Pohlig–Hellman, MOV/Frey–Rück, Smart/SSSA, and BSGS and rho in the generic group model, whose `Ω(√(εn))` lower bound (Shoup 1997) holds on every curve because the group is the same.

Proof. Tate: isogenous over `F_p` ⟺ same `#E(F_p)`. `n² > 16p > #E`, so the `n`-part is cyclic of order `n`. The embedding degree depends on `n` and `p` only. ∎

The prior study's Lean file `GenericInvariance.lean` proves the algorithm-level form for Nechaev-model algorithms. They have the same transcript on isomorphic groups. That model does **not** cover the hashed rho walks ecbench runs (sheriff case SHF-20261016-3caa6f). For those, equal cost is an empirical question, measured in `README.md`.

Theorem 2 — transfer, for every algorithm

Status: proved here; Lean: `TransferBound.transfer`, `DeepGap.transfer_comp_cost`, `DeepGap.chain_equivalence`.

Let `φ = φ_m ∘ … ∘ φ_1 : E → E'` be a chain of `F_p`-isogenies with `gcd(deg φ_i, n) = 1`, where `φ_i` costs `C_i` to evaluate at one point. Then for every always-correct algorithm `A'` on `E'` there is one on `E`, returning the same logarithm, with

`T(E) ≤ T_{A'}(E') + 2·Σ C_i` on every instance.

The solver may be anything: structural, reading the endomorphism ring, non-generic.

Proof.

1. `#ker φ_i = deg φ_i`; the isogenies are separable, since `p ∤ deg`.
2. So `ker φ ∩ ⟨P⟩` has order dividing `gcd(deg φ, n) = 1`.
3. Then `φ(P)` has order `n` and generates `E'(F_p)[n]`, and `φ(Q) = [k]φ(P)`.

4. Solve `(φ(P), φ(Q))` on `E'`: the answer is `k`. The cost is two evaluations of each `φ_i` plus the call.

Lean: composition preserves injectivity and the homomorphism laws, and the costs add (`Isogeny.comp`, `comp_cost`). Transfer along the composite costs `2·Σ C_i` and returns the same answer (`transfer_comp_cost`, `transfer_comp_answer`). With chains both ways, every solver on either end has a partner on the other (`chain_equivalence`). ∎

So **pairwise** hardness across a conductor gap is a question about the cost of the chain, nothing else.

Theorem 3 — the class measured here: explicit chains

Status: proved here; the arithmetic is in Lean. Chains cost `O(N + d)`.

Hypotheses:

- `t² − 4p = D₀·(2^d·N)²` with `h(D₀) = 1`, `D₀ ∉ {−3, −4}`, and 2 inert in `K` (`D₀ ≡ 5 (mod 8)`);
- `N` an odd prime, `N ∤ D₀`;
- **`t ≡ −2 (mod N)`**.

The class built by `volcano-gap` has `D₀ = −11`, `d = 6`, `N = 631409` (Lean: `class_discriminant`, `gap_prime_divides_t_plus_2`).

(a) `S1`: every `N`-kernel is twist-rational.

- Let `E ∈ S1`, so `N ∤ c` and `N | f/c = [O_c : Z[π]]`. Write `π = a + (f/c)·ω_c` with `O_c = Z[ω_c]`. Then `(π − a)/N ∈ O_c`, so π acts on `E[N]` as the scalar `a`.
- From `a² − ta + p ≡ 0` and `t² ≡ 4p (mod N)`, `a ≡ t/2 ≡ −1`.
- So the quadratic twist `E^t` has `E^t[N] ⊆ E^t(F_p)`, which is `(Z/N)²`. Lean `twist_order_sq_dvd`: `N² | 4·#E^t` from `4·#E^t = (t + 2)² − D₀f²`.
- Each of the `N + 1` cyclic `N`-subgroups of `E` has its `(N − 1)/2` abscissae in `F_p`. Vélu over `F_p` then builds the isogeny in `O(N)` field operations and evaluates it in `O(N)` per point.

(b) `S2`: one twist-rational line.

- Let `E ∈ S2`, so `N | c` and `N ∤ f/c`. Then π is not a scalar on `E[N]` (Kohel), so it is a Jordan block with eigenvalue −1.
- So `E^t(F_p)[N]` is one line. That line is the kernel of the unique ascending `N`-isogeny, which is built the same way in `O(N)`.

(c) The side certificate. `E ∈ S1 ⇔ E^t(F_p)[N] ≅ (Z/N)²`. In practice:

- S2 is proved by one witness point `R` with `[#E^t/N]R ≠ O`.
- S1 holds with error at most `N^{−16}` after 16 points that do vanish.

`volcano-gap certify` runs this on all 760 curve models.

(d) Shape.

- `N`-isogenies leave the 2-part of the conductor alone, and 2-isogenies leave the `N`-part alone.
- `h(O_K) = 1` and 2 is inert, so `S1` is a single 2-volcano with a one-vertex crater (`j(D₀)`) and `3·2^{a−1}` curves on level `a ≥ 1`: `3·2^d − 2` curves in all (Lean `volcanoSize_closed`; 190 for `d = 6`, `depth6_size`).
- `S2` is `h(O_N) = N − (D₀/N)` volcanoes of the same shape.
- In this class `(−11/N) = (N/11) = +1` (Lean `gap_prime_mod_11`), so `h(O_N) = N − 1`. Two of the `N + 1` `N`-isogenies from `j(D₀)` are horizontal (endomorphisms) and `N − 1` descend, one to each `S2` crater.
- Each 2-edge is a Vélu 2-isogeny, and its dual is again a 2-isogeny edge of the volcano.

(e) Floor no harder than crater, explicitly. Take `E ∈ S2` at level `a` and `E' ∈ S1`:

1. ascend by E's twist-rational line, one N-isogeny into S1 at level a;
2. walk at most 2d 2-isogenies to E'.

By Theorem 2, T(E) ≤ T_{A}(E') + O(N + d) per instance, for every A. The construction is O(N + d + log p).

(f) Crater no harder than a given floor curve. Take E ∈ S1 and a given E' ∈ S2:

1. E' ascends to some E_1 ∈ S1, and the dual E_1 → E' has a twist-rational kernel;
2. finding that kernel among E_1's N + 1 lines costs $O(N^2)$ once, or $\tilde{O}(N)$ with Bostan–Morain–Salvy–Schost (literature; not implemented here);
3. after that, T(E) ≤ T_{A}(E') + O(N + d) per instance.

(g) No KKM isolation in this class.

- From $j(D_0)$, a uniformly random line of E[N] is horizontal with probability 2/(N + 1). Otherwise it lands on a uniformly random S2 crater.
- A uniformly random descending 2-walk then lands on a uniform curve of any chosen level.
- So if some algorithm solves a fraction ε of the S2 curves of one level, it solves the S1 crater in expected O((N + d)/ε) field operations plus O(1/ε) calls.

The isolation Koblitz, Koblitz and Menezes (2011) proposed for crater curves rests on Problem B, descending from a given crater curve. Problem B is trivial in this class because t ≡ −2 (mod N). The same argument works with O(N·k)-size extension arithmetic whenever t/2 mod N has small order k in F_N^×/±1.

Cost measured here (run/pair.json, volcano-gap construct):

step	measured
build the descending N-isogeny	0.15 s, about $4.4 \cdot 10^7$ field multiplications
evaluate it at one point	about $3.2 \cdot 10^6$ field multiplications, 0.019 s
transfer one instance (2 points)	about $6.3 \cdot 10^6$ field multiplications, 0.04 s
one rho.negation solve, for comparison	about $1.29 \cdot 10^6$ group operations, 0.39 s mean wall time (README.md)

∎

Theorem 4 — where "the same" fails: automorphisms

Status: proved here (a constant factor); measured in this repository.

Aut(E) = O_{c(E)}^×, which is {±1} unless O_c = Z[i] (order 4) or Z[ζ₃] (order 6). Those are maximal orders, so c = 1.

So in a class with D_0 ∈ {−3, −4}:

- exactly the crater curves with c = 1 (j = 1728 or 0) have extra automorphisms, and they lie in S1;
- a rho walk on classes of size |Aut| costs about √(πn/(2|Aut|)), a factor √3 or √2 below negation rho (Duursma–Gaudry–Morain 1999).

Measured in this repository:

- rho.cubic against rho.negation read 1.69–1.92 against √3 = 1.73 (drastic round 2, round 3);
- the √2 walk under-realizes, at 1.14–1.32.

This is a **constant factor on one method run where the curve is given**. It is not a difference in T*:

- by Theorem 2 and the cheap ascent (Theorem 3(e) here; Galbraith 2024 in general), a floor curve's instance can be carried to the j = 0 crater and solved there with the cubic walk;
- so T*(floor) ≤ T*(crater) + transfer;
- and T*(crater) ≤ T*(floor) + transfer by Theorem 2 with the descent.

The √3 follows the instance once the map is built. In this study's class $D_0 = -11$, so Aut = ±1 on every curve and there is no exception.

Theorem 5 — general classes

Status: literature, as reported; not re-derived here.

Galbraith, *Climbing and descending tall isogeny volcanos*, ePrint 2024/924 (summary: crypto-autoresearcher/knowledge/literature/KN-LIT-ca35e0.md):

- given two curves of a class with q^{1/4} < N < q^{1/2} joined by an N-isogeny, a representation is found in Õ(N^{1/2}) = Õ(q^{1/4}) (Kani/Robert);
- ascending from any floor curve costs at most Õ(q^{1/4});
- for tall volcanoes with polynomial-size craters, which include CM/class-number-one classes like this one, an isogeny between **any two given curves** costs Õ(q^{1/4}), rigorously;
- in general classes, heuristically Õ(q^{2/5}).

With Theorem 2 this gives, for any two **given** curves of a class: |T*(E) − T*(E')| ≤ Õ(q^{1/4}) (small crater) or Õ(q^{2/5}) (general). Both are o(√q). So the generic exponent 1/2, and any sub-√q exponent above 1/4, is shared by every curve of the class, gap or no gap.

Open — what cannot be claimed

Problem B / the KKM speculation. Is there a weakness held by a positive fraction of S2 curves that a given S1 curve cannot reach faster than √q?

- Reaching it means producing S2 curves from S1. The best known cost is min(N², q^{1/2}) when t/2 mod N has large order (Galbraith 2024 §11, stated as the main open problem).
- In such classes the random self-reduction (Jao–Miller–Venkatesan 2005; Galbraith Thm 5) covers S2 but not S1.
- Theorem 3(g) closes it only for the special classes with t/2 of small order mod N, like the one measured here.
- No proof, and no measurement at toy size, settles it in general. KKM's argument assumes the weakness is independent of the endomorphism ring, and no such weakness is known on prime fields.

Exact equality of one method's constant. Theorems 1–2 bound the best cost T*. They do not say that a particular hashed walk (ecbench's rho.negation) has the same mean on every curve. That is what the measurement in README.md tests, within ±5%, at 52 bits.

Proof map

Declarations are in namespace DeepGap unless marked.

- **Thm 2, one edge.**
  - Lean: TransferBound.transfer, TransferBound.transfer_cost, TransferBound.transfer_answer.
  - Hypotheses: a Hom (injective, preserves one and mul) and an always-correct Solver.
  - Status: proved in the 2026-10-10 study.

- **Thm 2, chains.**
  - Lean: `Isogeny.comp`, `comp_cost`, `transfer_comp_cost`, `transfer_comp_answer`, `chain_equivalence`.
  - Hypotheses: the same.
  - Status: proved.
- **Thm 3(a), the arithmetic.**
  - Lean: `twist_order_identity`, `twist_order_sq_dvd`, `order_mod`.
  - Hypotheses: integers `p, t, D, f, N` with `t² − 4p = Df²`, `N | f` and `N | t + 2`.
  - Status: proved.
- **Thm 3, the class.**
  - Lean: `class_discriminant`, `gap_prime_divides_t_plus_2`, `class_twist_order`, `gap_prime_mod_11`.
  - Hypotheses: the numbers of `run/pair.json`.
  - Status: proved by `decide` in the kernel.
- **Thm 3(d), the counts.**
  - Lean: `volcanoSize_closed`, `depth6_size`.
  - Hypotheses: the level-count recursion `1, 3·2^{k−1}`.
  - Status: proved. The recursion itself (the class-number formula for inert 2) is not formalized.
- **Thm 1, the curve parts of Thm 3(a)–(c), and Thm 4.**
  - Lean: none. They need elliptic curves over `F_p`, `End`, Vélu and the Weil pairing.
  - Status: not formalizable against today's Mathlib. `volcano-gap` checks them numerically on every curve.

Axiom audit: `lean/axiom-audit.log`. Every declaration uses at most `propext`, `Classical.choice` and `Quot.sound`. `tools/lean/proof-gate.sh` passes the project: 18 theorems, 0 trusted.